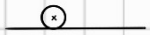
Corpo rigido in rotazione intorno a un asse fisso

$\tau = I\ddot{\theta} = I\frac{d^2\theta}{dt^2}$ $\quad \tau = \sum_i R_i F_i \sin\theta_i$ (massa)



$T = \frac{1}{2} I \omega^2$ $\quad I = \sum_i m_i R_i^2$

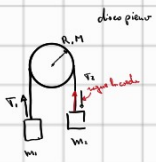
moto di rotolamento



$N_{cm} = \omega R$ $\quad T = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I \omega^2$

- DINAMICA: -> serve l'attrito (statico)

Macchina di Atwood



$$\begin{cases} m_1 a = m_1 g - T_1 \\ m_2 a = -m_2 g + T_2 \\ I\alpha = T_1 R - T_2 R \end{cases}$$